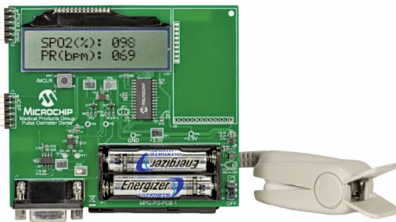


Nidhi Agarwal is Senior Technology Journalist at EFY with a deep interest in embedded systems, development boards, and IoT cloud solutions.

### Advanced Pulse Oximeter



The reference design supports a low-cost, low-power pulse oximeter for measuring heart rate and oxygen levels, integrating signal processing, power management, and data handling while reducing the need for extra components.

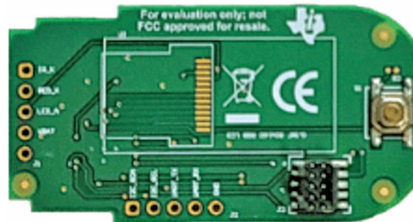
Microchip's pulse oximeter reference design demo board enables the development of a low-cost, low-power handheld or wearable pulse oximeter with a user interface for measuring heart rate and blood oxygen levels. It serves as a foundation for hospital or clinical pulse oximeter designs and wearable activity trackers. The device measures heart rate and oxygen saturation using a dsPIC33FJ128GP802 with a 513th-order digital-FIR bandpass filter, eliminating the need for an analogue front-end and reducing costs. A pulse-width modulation module drives the LCD, while an MCP1640 boost regulator, MCP4728 DAC, and MCP6002 dual op-amp manage power and signal conditioning. It features an 8-channel hardware direct memory access (DMA) system with a 2kbyte dual-ported DMA buffer. The design includes power management with an on-chip 2.5V voltage regulator and real-time clock source switching for performance.

The reference design is suitable for wearable and clinical applications.

Microchip

<https://www.electronicsforu.com/electronics-projects/single-chip-pulse-oximeter-reference-design-with-90db-dynamic-range-for-lower-perfusion-index-2>

### Single-Chip Pulse Oximeter



The reference design enables a single-chip pulse oximeter with signal processing, power management, and data handling. It supports heart rate and oxygen measurement, low power consumption, and integration into patient monitoring systems.

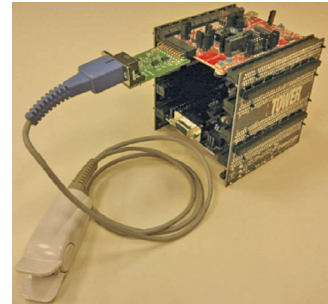
Texas Instruments' TIDA-010267 reference design demonstrates the analogue capabilities of the MSPM0L1306 in a single-chip pulse oximeter. The MSPM0L1306 uses its operational amplifiers as a trans-impedance amplifier and current control driver with zero-drift and low-noise properties. An integrated analogue-to-digital converter (ADC) enables oversampling, achieving a 90dB dynamic range. The design includes a graphical user interface that displays photoplethysmography (PPG) waveforms and provides heart rate and oxygen saturation measurements. It supports heart rates from 30 to 240 BPM and a perfusion index range of 0.1% to 20%. The internal 12-bit ADC ensures readings with a display resolution of 1 BPM for heart rate and 1% for SpO2. The device supports battery life with low power consumption and a shutdown current below 83nA. It is used in clinical pulse oximeters and multiparameter patient monitoring systems, contributing to the assessment of oxygen saturation and patient care.

The reference design is suitable for clinical applications.

Texas Instruments (TI)

<https://www.electronicsforu.com/electronics-projects/single-chip-pulse-oximeter-reference-design-with-90db-dynamic-range-for-lower-perfusion-index>

### Tower System Module Pulse Oximeter



The reference design supports pulse oximetry and medical prototyping with microcontrollers, development tools, and sensor integration. It enables measurement, connectivity, and system expansion for various applications within a development platform.

The MED-SP02 tower system module reference design is a medical development tool for the 9S08MM Flexis 8-bit, MCF51MM ColdFire 32-bit, and Kinetis 32-bit MCU families, supporting pulse oximetry application development and compatibility with multiple tower system modules. It measures oxygen levels using two light wavelengths, 660nm (red) and 940nm (infrared), based on differences in absorption by deoxygenated and oxygenated haemoglobin. The 9S08MM128/64/32 is an 8-bit microcontroller with USB connectivity, graphic display support, and measurement accuracy for medical and industrial applications. The MCF51MM256/128 is a 32-bit microcontroller with low power operation, USB connectivity, and analogue performance for medical devices and industrial control. The TWR-K53N512 supports product development for the Kinetis MCU family, functioning as a standalone debug tool and integrating with the Tower System. The TWR-MCF51MM module supports the MCF51MM256 Flexis ColdFire 32-bit MCUs for medical prototyping, including an electrocardiograph sensor.

The reference design is suitable for medical and industrial applications.

NXP Semiconductors

<https://www.electronicsforu.com/electronics-projects/pulse-oximeter-reference-design-2>